

QUICK CHALLENGE 🔥

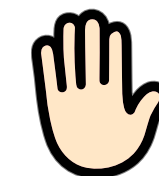
Your task is to calculate the total number of **weights** and **biases** in a CNN with the following architecture:

1. Input: $64 \times 64 \times 3$ RGB image
2. Conv1: 16 filters, Kernel size: 3×3 , Stride: 1, Padding: 1
3. Average Pool 1: 2×2 , Stride: 2
4. Conv2: 32 filters, Kernel size: 3×3 , Stride: 1, Padding: 1
5. Average Pool 2: 2×2 , Stride: 2
6. Conv3: 64 filters, Kernel size: 3×3 , Stride: 1, Padding: 1
7. Fully Connected 1: 256 neurons
8. Fully Connected 2: 128 neurons
9. Output Layer: 10 classes

Instructions:



You have 10 mins



Raise your hand
when you are done



Write your answer
on the board

Those who used
gpt to solve this,
idk what you will
do in the exam :)



SOLUTION

Output Shapes

- Input:
 $64 \times 64 \times 3$
- Conv1 (16 filters, 3×3 , padding=1):
 $64 \times 64 \times 16$
- AvgPool1 (2×2 , stride=2):
 $32 \times 32 \times 16$
- Conv2 (32 filters, 3×3 , padding=1):
 $32 \times 32 \times 32$
- AvgPool2 (2×2 , stride=2):
 $16 \times 16 \times 32$
- Conv3 (64 filters, 3×3 , padding=1):
 $16 \times 16 \times 64$
- Flatten:
 $16 \times 16 \times 64 = 16,384$

Parameter Calculation

- | | |
|--|--------------------------------------|
| 1. Conv1 ($3 \rightarrow 16$): | 4. FC1 ($16,384 \rightarrow 256$): |
| • Weights: $3 \times 3 \times 3 \times 16 = 432$ | • Weights: 4,194,304 |
| • Biases: 16 | • Biases: 256 |
| • Total: 448 | • Total: 4,194,560 |
| 2. Conv2 ($16 \rightarrow 32$): | 5. FC2 ($256 \rightarrow 128$): |
| • Weights: $3 \times 3 \times 16 \times 32 = 4,608$ | • Total: 32,896 |
| • Biases: 32 | |
| • Total: 4,640 | |
| 3. Conv3 ($32 \rightarrow 64$): | 6. Output ($128 \rightarrow 10$): |
| • Weights: $3 \times 3 \times 32 \times 64 = 18,432$ | • Total: 1,290 |
| • Biases: 64 | |
| • Total: 18,496 | |

$$448 + 4640 + 18,496 + 4,194,560 + 32,896 + 1,290 = 4,252,330$$